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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R-26-IT-021

1. Topic (12 words max)

Intelligent Ayurvedic Herbal Plant Identification And Economic Analysis In Sri Lanka

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The agriculture industry plays an important role in the development of Sri Lanka's economy and food security situation; however, most of its small farmers still base their irrigation and fertilization practices on manual experience and traditional knowledge. As stated by the Food and Agriculture Organization (FAO), the variations of the climate and poor management of resources negatively affect agricultural efficiency and crop losses [1].

Current Approaches: The current farming approaches use either manual farm inspection or stand-alone weather applications that only generate generic forecasts. These systems do not incorporate real weather conditions, moisture and acidity level of the soil, wind exposure, and light intensity into one platform.

Social Impact: Over two billion people in the world rely on agriculture to earn their living. The incorrect decisions that are made in agriculture as a result of changed weather conditions and ineffective soil monitoring may lead to decreased crop production, food insecurity, and losses for farmers, especially in developing countries like Sri Lanka [3].

Economic Impact: Over-irrigation, ineffective fertilization, and delayed reaction to changing weather conditions make the cost of production higher, decrease the quality of crops, and yield. Data-driven precision agriculture technologies have proven to be more efficient and less costly [4].

Technical Impact: Despite IoT sensors and weather APIs being freely available, there are not enough agricultural systems that integrate all these technologies, provide analytics in real time, work automatically, and have user-friendly interface. Many farmers have to use several different tools to get the needed environmental data [5].

Environmental Impacts: Water overuse and overuse of fertilizers are responsible for soil degradation, water contamination, and green house gases emissions. Tracking the soil moisture content and soil pH along with weather conditions helps achieve sustainable agriculture practices through proper irrigation and fertilization [6].

The objective of this paper is to propose an IoT based Smart Agriculture Monitoring System which can integrate the real time data of weather, automated location detection, soil moisture content, soil pH, wind exposure, and sun light together.

References

[1] Food and Agriculture Organization of the United Nations. (2022). The State of Food and Agriculture 2022. FAO.

[2] Wolfert, S., Ge, L., Verdouw, C., & Bogaardt, M. J. (2017). Big Data in Smart Farming: A review. Agricultural Systems, 153, 69–80.

[3] World Bank. (2023). Agriculture and Food. World Bank.

[4] Zhang, C., & Kovacs, J. M. (2012). The application of small unmanned aerial systems for precision agriculture. Precision Agriculture, 13(6), 693–712.

[5] Ray, P. P. (2017). Internet of Things for Smart Agriculture: Technologies, Practices and Future Direction. Journal of Ambient Intelligence and Smart Environments, 9(4), 395–420.

[6] United Nations. (2015). Transforming Our World: The 2030 Agenda for Sustainable Development (SDGs 2, 12 & 13).

6. Brief description of Existing Research and Systems (300 words max)

Existing research on Sri Lankan Ayurvedic medicinal plants has mainly addressed species identification, quality assessment, knowledge representation, and cultivation support. However, these systems are largely fragmented and suffer from several limitations [1].

In plant identification, researchers have developed CNN-based models such as EfficientNet-B0 and VGG-16 for Ayurvedic plant classification, achieving high accuracy (92–99%) [2,3]. Mobile applications using image recognition also exist, but most lack focus on Sri Lankan endemic species and integrated Ayurvedic knowledge [4].

For quality assessment, deep learning models have been applied for disease detection in medicinal plants. However, the majority function as black-box systems, limiting their trustworthiness in medical applications [5]. Recent studies have started incorporating Explainable AI using Grad-CAM with ResNet models to highlight biologically relevant features such as lesions and maturity signs [6,7].

Knowledge preservation has been attempted through ontology-based systems [8] and basic chatbots. These approaches often require technical expertise (e.g., SPARQL queries) or rely on shallow pattern matching, failing to deliver in-depth domain knowledge [1]. Cultivation recommendation systems are mostly general and lack integration of soil, climate, market demand, and economic factors specifically for small-scale Ayurvedic herbal farmers [9].

Major gaps include poor interpretability, limited real-world robustness, absence of integrated platforms, and insufficient domain-specific solutions for Sri Lankan Ayurveda [1,5,8].

The proposed integrated project addresses these gaps by developing a comprehensive ecosystem combining mobile detection, explainable quality assessment, knowledge intelligence, and cultivation recommendations.

References

[1] Jeyananthan et al., “Ontology for Sri Lankan Medicinal Plants,” Proceedings of the International Conference on Information Technology, Sri Lanka, 2020.

[2] Lankasena et al., “Misidentifications in Ayurvedic Medicinal Plants and CNN-based Identification,” Computers in Biology and Medicine, 2024.

[3] Jayanka and Fernando (2020); Pathiraneg et al.

[4] Lee et al., “Deep-Plant: Plant Identification with Convolutional Neural Networks,” 2015. https://arxiv.org/abs/1506.08425

[5] Haque et al. (2025); Gamage et al. (2024)

[6] Gopalan et al., “Corn leaf disease diagnosis: enhancing accuracy with ResNet152 and Grad-CAM,” 2025.

[7] Yadav et al., "A Hybrid Deep Learning Framework for Medicinal Plant Species Detection with Explainable AI and GradCAM Visualization."

[8] S. Jeyananthan et al., Ontology-based system for Sri Lankan medicinal plants.

[9] D. G. N. D. Gamage et al., "Ethnopharmacological Survey on Medicinal Plants Used in Sri Lanka," 2021. https://pmc.ncbi.nlm.nih.gov/articles/PMC8257331/

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an integrated intelligent ecosystem titled "Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka". It combines four tightly coupled modules that work together to support plant identification, quality assessment, traditional knowledge access, and cultivation decision-making for Ayurvedic medicinal plants.

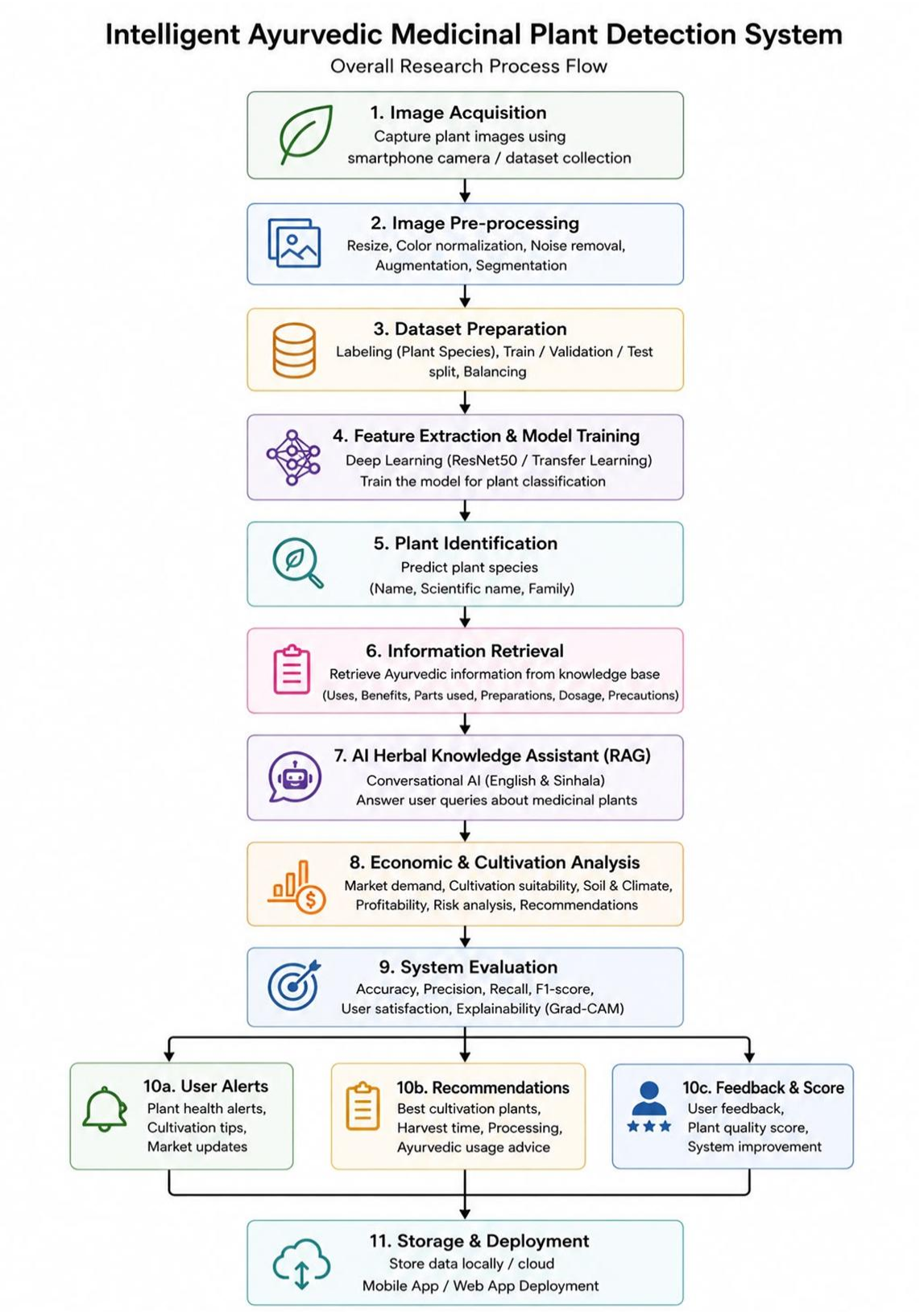
Solution Nature The system follows a three-tier architecture consisting of a Mobile/Web Frontend, a Backend Processing Layer, and an AI Services Layer. Users interact through a Flutter-based mobile application that allows image capture/upload, natural-language queries, and cultivation recommendation requests. The backend (Flask/FastAPI) handles image preprocessing, API routing, authentication, and data storage. The AI layer contains four specialized engines that share a common medicinal plant knowledge base and image dataset.

Data Flow

1. User captures or uploads an image of a plant leaf/flower/fruit.
2. The image is preprocessed (resized, normalized) and sent to the Plant Detection module for species identification.
3. Parallelly, the Quality Assessment module (ResNet50 + Grad-CAM) classifies the plant material as healthy, diseased or overmature and generates a visual heatmap.
4. Identified plant name is used to retrieve detailed Ayurvedic information from the RAG-based Knowledge Assistant (English + Sinhala).
5. For farmers, the Cultivation Recommendation Engine combines soil type, climate data, land size, market demand and cost factors to suggest suitable plants, expected yield and economic viability.
6. Results (plant name, quality status, heatmap, medicinal uses, and cultivation advice) are returned to the mobile interface.

The system is designed to be modular, scalable and partially edge-capable so that basic identification and quality checks can function with limited internet connectivity. All modules share a verified knowledge base and a carefully curated image dataset of Sri Lankan Ayurvedic plants, ensuring consistency and domain relevance across the ecosystem.

Conceptual Diagram The high-level architecture can be visualized as follows:



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Several domains of expertise are required for this project, such as Ayurvedic medicinal plant identification, image processing, deep learning, explainable artificial intelligence, natural language processing, recommendation systems, and agricultural data analysis. The project team will conduct research on academic papers related to medicinal plants in Sri Lanka, AI-based medicinal plant identification, medicinal plant quality assessment, plant cultivation recommendation systems, and AI-based knowledge retrieval. Existing studies indicate that medicinal plant identification often relies on expert knowledge and that automated systems can improve accessibility and identification accuracy, especially for non-expert users.

The project also incorporates an AI-based cultivation recommendation module that considers soil conditions, climate data, land size, market demand, expected income, and potential risks when recommending appropriate Ayurvedic plants for cultivation. Technical skills needed for implementation include mobile application development, backend API development, database management, machine learning model training, image processing, model evaluation, and AI integration. The plant identification component will require computer vision and transfer learning skills to classify medicinal plants from images.

The quality assessment component will require skills in Python, PyTorch, OpenCV, ResNet50 transfer learning, and Grad-CAM visualization to classify plant materials as healthy, diseased, or overmature while explaining the decision made by the model. The herbal knowledge assistant component will require expertise in Retrieval-Augmented Generation, multilingual semantic search, vector databases, LangChain, ChromaDB, Flask backend development, and Flutter-based frontend integration. The project relies on several data sources. Image data will be required for medicinal plant species identification and plant quality assessment. These images may include leaves, flowers, fruits, roots, and other plant parts collected through field photography, herbal gardens, farms, Ayurvedic institutions, or publicly available plant image datasets.

The quality assessment dataset will be labelled using conditions such as healthy, diseased, and overmature. The cultivation recommendation component will require environmental, agricultural, and economic information such as soil type, climate conditions, land size, market demand, production potential, and income-related data. The herbal knowledge assistant will require a verified knowledge base containing plant names, descriptions, benefits, parts used, Ayurvedic applications, and other traditional medicinal information in both English and Sinhala. The ethical implications of this project are significant, as it deals with medicinal information, user queries, agricultural recommendations, and AI-based image predictions.

The system should clearly state that it is a decision-support system and is not intended to replace qualified Ayurvedic practitioners or domain experts, since incorrect plant identification or quality assessment may lead to the misuse of medicinal plants. All data gathered from users, farms, or institutions should be used only with permission, and personal or location-related information should be protected. The system should not provide unsafe medical advice and should present medicinal information in an educational manner. Image datasets should be gathered and documented responsibly, with appropriate credit given to the sources. The AI models should also be tested for accuracy, bias, and reliability before deployment to ensure that users can trust the system's outputs safely.

9. Objectives and Novelty

Main Objective The aim of this project is to develop an integrated intelligent ecosystem for Ayurvedic herbal plants in Sri Lanka titled "Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka". The system will combine mobile-based plant detection, explainable quality assessment, AI-powered knowledge intelligence, and cultivation recommendations to preserve traditional knowledge, ensure quality in the herbal supply chain, support small-scale farmers, and promote sustainable Ayurvedic practices.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22640598	Herbal Knowledge Intelligence System	• Develop a Retrieval-Augmented Generation (RAG) based conversational AI supporting English and Sinhala. • Build a knowledge base of 20-30+ Sri Lankan medicinal plants. • Integrate LangChain, ChromaDB, and LLM for natural language queries. • Develop Flutter mobile frontend with Flask backend.	RAG-powered multilingual (Sinhala + English) conversational AI assistant specifically designed for Sri Lankan Ayurvedic medicinal plants, overcoming limitations of rigid ontologies and shallow chatbots.
IT22132796	AI-Driven Cultivation Recommendation System For Small-	• Collect and integrate data on soil conditions, climate, plant characteristics, market prices, demand, and cultivation costs.	AI system that integrates agronomic, environmental, and economic factors to provide practical cultivation recommendations specifically tailored

	Scale Herbal Farmers	• Develop AI models for personalized cultivation recommendations, yield prediction, and economic analysis. • Build user-friendly interface for small-scale farmers.	for small-scale Ayurvedic herbal farmers in Sri Lanka.
IT22168986	Mobile-Based Detection Of Sri Lankan Ayurvedic Plants Using Image Recognition	• Collect dataset of Sri Lankan Ayurvedic plant leaves and flowers. • Train CNN/deep learning model for accurate plant identification. • Develop mobile application for real-time image scanning. • Integrate identification results with medicinal information database.	A dedicated mobile application focused on Sri Lankan Ayurvedic plants using image recognition technology, providing instant identification along with Ayurvedic medicinal uses and benefits.
IT22112132	Explainable AI System For Ayurvedic Medicinal Plant Quality Assessment	• Develop ResNet50 model using transfer learning for classifying plant materials (healthy, diseased, overmature). • Integrate Grad-CAM to generate visual explanations (heatmaps). • Create a labeled dataset and evaluate model performance. • Develop user interface for image upload and heatmap visualization	Introduces Explainable AI (XAI) using Grad-CAM specifically for quality assessment of Ayurvedic medicinal plants in Sri Lanka, addressing the critical transparency and trust issues in black-box deep learning models for herbal medicine.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	X	No	

b) The proposed topic exhibits novelty.

Yes	X	No	

c) The student group can successfully execute the proposed project.

Yes	X	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	X	No	

e) Any other comments:

Highly recommended

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Nalum	Amarasena	12/01/2026
Co-Supervisor	Ms.	Kanchika	Kahatapitiya	12/01/2026
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				